Efficient Training and Real-Time Inference for Embodied Foundation Models

From Transformer Optimization to Closed-Loop VLA / WAM Robot Execution

Foundation of Embodied AI ·Xiong Zeng and contributors

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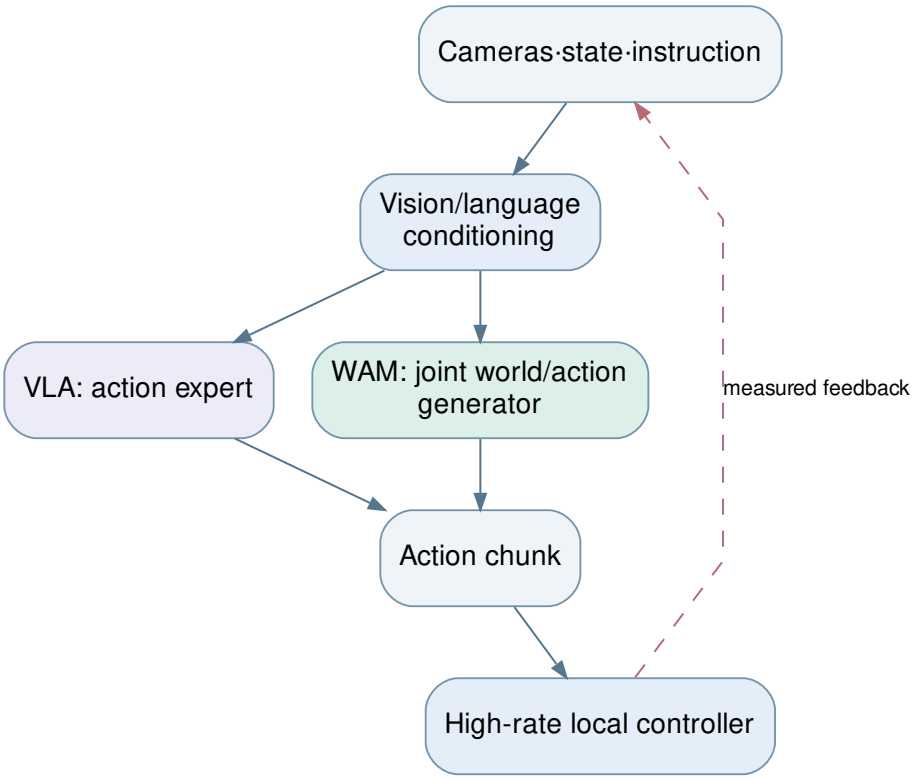
1. Begin with Three Optimization Objectives

> **Scope.** This lecture connects Transformers, flow matching, vision-language-action models (VLAs), world action models (WAMs), and robot control. It addresses three questions: how to train with less compute, how to generate actions at low latency, and how to keep a physical robot moving continuously while a relatively slow VLA or WAM runs. Full WAM architectures, RL post-training algorithms, code, and hands-on experiments are outside its scope.

Embodied foundation models commonly contain vision encoders, language and temporal backbones, and action-generation modules. A VLA primarily produces actions from observations and instructions. Some WAMs additionally predict future video, geometric states, or compact latent world representations, aligning these predictions with the actions. **Using a Transformer and predicting the future world are distinct architectural choices.**

Setting	Primary constraint	Metrics that matter
Pretraining and fine-tuning	Compute, VRAM, data throughput	GPU-hours, steps/s, peak VRAM, validation-task quality
VLA / WAM inference	Prefix processing, iterative generation, visual tokens	Per-request p50/p95/p99, VRAM, action-chunk latency
Robot closed-loop execution	Observation freshness, jitter, motion continuity	Observation-to-action delay, action-buffer slack, tracking error
High-frequency servo loop	Feedback, contact, safety	Control rate, force/pose tracking, deadline and safety limits

Figure 1: VLA / WAM inference and closed-loop control (conceptual)



Fast servo control does not require equally fast foundation-model inference. A practical hierarchy periodically generates action chunks in the VLA or WAM, while a separate position, velocity, torque, or impedance controller executes and corrects them at a higher rate. Model request rate, action horizon, low-level servo frequency, and actual feedback frequency must be distinguished.

An approximate request-latency breakdown is

$$T_{\text{req}} \approx T_{\text{camera}} + T_{\text{pre}} + T_{\text{encoder}} + T_{\text{context}} + T_{\text{generator}} + T_{\text{transfer}} + T_{\text{queue}}.$$

This is a **profiling decomposition**, not a strict serial identity: stages can overlap. Accelerating an individual kernel will not necessarily produce the same end-to-end speedup. For a WAM, $T_{\text{generator}}$ may include joint video-action denoising; decoding pixels for an external display is an optional extra step, depending on the deployment.

2. Why Are Transformers and WAMs Expensive?

2.1 Attention Compute and Memory

For $X \in \mathbb{R}^{L\times d}$, scaled dot-product attention is

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \qquad \text{Attn}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V.$$

M denotes a causal, modality, or valid-token mask. Standard dense attention has approximately $O(L^2 d)$ arithmetic com-

plexity; materializing the attention matrix requires $O(L^2)$ intermediate storage. Multi-camera high-resolution observations, video history, and future-video tokens increase L. Long-horizon WAMs may therefore be more exposed to attention and activation-memory costs, although not every WAM is inherently slower than a VLA.

2.2 Video Tokens Introduce an Additional Cost Dimension

Suppose a latent-video representation contains F frames, each spatially encoded into $H' \times W'$ positions after VAE compression and patchification. A useful scaling approximation is

$$L_{\text{video}} \propto F\,H'W', \qquad L_{\text{total}} = L_{\text{context}} + L_{\text{future}} + L_{\text{action}}.$$

Even when future RGB frames are never displayed, future-video latent tokens can still participate in every iteration of the joint video-action DiT. Increasing frame count, camera count, or resolution may raise training activation memory, attention compute, and inference time. Excessive compression, however, can remove contacts, small clearances, or rapid slip. **Minimizing token count is not equivalent to optimizing robotic competence.**

2.3 Training Memory and Input Pipelines

$$M_{\text{train}} \approx M_{\text{params}} + M_{\text{grads}} + M_{\text{optim}} + M_{\text{act}} + M_{\text{temp}}.$$

Weights, gradients, optimizer state, retained activations, and temporary tensors should be tracked separately. A quantized checkpoint's file size is not the VRAM needed for full-parameter training. Video policies also pay for multi-view decoding, VAE encoding, frame-rate alignment, and action-image timestamp synchronization. **When the GPU is underutilized, inspect CPU decoding, storage/network throughput, and augmentations before adding more accelerators.**

3. FlashAttention: Efficient Exact Attention

3.1 The Problem Is Intermediate Matrices and HBM Traffic

For one attention head, the naive implementation forms

$$S = QK^\top/\sqrt{d}, \qquad P = \text{softmax}(S), \qquad O = PV.$$

Although O has only $L \times d$ elements, both S and P contain L^2 entries. For $L = 4096$ with BF16 (2 bytes per element), **one** explicit 4096×4096 matrix for **one head** occupies approximately 32 MiB, excluding other heads, gradients, and workspace.

3.2 Tiling and Online Softmax

FlashAttention tiles Q, K, V, keeping stable softmax statistics on chip instead of materializing full S and P in GPU high-bandwidth memory (HBM). For one query row, suppose the running maximum, exponential normalizer, and unnormalized value sum are (m, ℓ, u). For new logits s_j and values v_j, update

$$\begin{aligned} m' &= \max(m, \max_j s_j), \\ \ell' &= e^{m-m'}\ell + \sum_j e^{s_j - m'}, \\ u' &= e^{m-m'}u + \sum_j e^{s_j - m'} v_j, \\ o &= u'/\ell' \quad \text{(after all tiles).} \end{aligned}$$

The result is **exact softmax attention up to floating-point rounding**, not an approximation obtained by discarding tokens. Standard dense-attention arithmetic remains $O(L^2 d)$. FlashAttention-2 improves work partitioning; PyTorch `scaled_dot_product_attention` (SDPA) can dispatch to an optimized implementation when supported.[4,5,10]

Technique	What is reused or reduced?	Approximate?
FlashAttention	HBM traffic and intermediates within one attention pass	No (apart from rounding)
KV cache	K/V of **unchanged context** across computations	No, if dependencies truly permit reuse
DiT caching	Velocity or features from neighboring denoising steps	Usually yes
Few-step generation / distillation	Calls to the iterative generative model	Typically changes the model or trajectory

Implication for WAMs. First profile attention-layer time and memory. Only then decide whether SDPA/FlashAttention, token compression, or fewer joint-DiT iterations provides the greatest benefit. FlashAttention cannot by itself remove repeated forward passes through a multi-billion-parameter DiT.

4. Efficient Training: VLA versus WAM Bottlenecks

Technique	Main benefit	Important limitation
BF16 / mixed precision	Lower compute and some memory costs	Requires numerical and device checks
FlashAttention / SDPA	Less attention memory traffic for long sequences	Depends on masks, length, and hardware
Activation checkpointing	Trades recomputation for activation VRAM	Backward can be slower
LoRA / freezing modules	Reduces trainable state and optimizer memory	Full backbone forward may remain expensive
FSDP / parameter sharding	Shares parameters and optimizer states across GPUs	Communication and synchronization overhead
Cached VAE features	Avoids repeated encoding when encoder is frozen	Invalid when augmentations/temporal sampling change meanings
Video tokens / temporal subsampling	Shortens attention context	May lose fine motion and contact alignment
Prefetch / asynchronous decoding	Prevents data-loading stalls	Consumes CPU, RAM, and storage bandwidth

4.1 VLA: Separate the Visual Prefix from the Action Expert

For $\pi0/\pi0.5$-style fine-tuning, the vision encoder, VLM backbone, and action expert need not use the same update policy. Freezing or LoRA reduces trainable state, but gradient-bearing computation through a large backbone can still retain substantial activation memory. In online off-policy RL, actor inference, replay decoding, and learner updates are distinct bottlenecks; these are post-training infrastructure issues, not necessarily attention-kernel issues.[1,7]

4.2 WAM: Account for Video Compute Explicitly

A simplified way to express joint video-action training is

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{video}} + \lambda_a \mathcal{L}_{\text{action}}.$$

This is **only a conceptual cost decomposition**; not every WAM literally sums two independently defined losses. DreamZero, for example, trains a weighted joint flow-matching velocity field.[11] Such models should profile video VAE encoding, spatiotemporal DiT layers, action heads, sequence length, and gradient flow separately. Cache video features

only if the corresponding encoder is frozen and preprocessing semantics stay fixed. Next consider prediction horizon, latent resolution, and trainable-module selection.

Contact-rich tasks impose a constraint. In insertion, tool rotation, aggressive video subsampling or latent compression may hide collision edges, initial contact, and slip. GPU-hours alone are not sufficient: action error, contact continuity, and real-task success still matter.

5. KV Cache and DiT Caching: When Reuse Is Valid

5.1 KV Memory and Conditions for Correct Reuse

For Transformer layer ℓ, cached past keys and values can be written as

$$\mathcal{C}_t^{(\ell)} = (K_{1:t}^{(\ell)}, V_{1:t}^{(\ell)}), \qquad M_{\mathrm{KV}} \approx 2 n_\ell B L h_{\mathrm{kv}} d_h b.$$

Here n_ℓ is the number of layers, B the batch size, L the context length, h_{kv} the number of KV heads, d_h each head dimension, and b bytes per element. For 32 layers, eight KV heads, dimension 128, BF16, batch one, and 2,048 tokens, the cache alone is about 256 MiB, excluding weights.[1]

A cache is **exactly reusable** only if cached hidden representations do not depend on subsequently changing tokens, and the weights, positions, masks, and input semantics remain compatible. With fixed prefix p and changing generative suffix g, a conceptual dependency mask is

$$M = \begin{bmatrix} M_{pp} & -\infty \\ M_{gp} & M_{gg} \end{bmatrix} \quad \text{(rows: queries; columns: keys).}$$

The upper-right block prevents the **prefix from attending to the mutable suffix**. This is an illustrative mask, not a universal design for all Transformer variants.

5.2 π0 / π0.5: Fixed Conditioning Prefix and Mutable Actions

Let $c = (I, l, s)$ contain image, instruction, and robot state. Under the convention "$\tau = 0$ is clean, $\tau = 1$ is noise," a flow-matching path is

$$x_\tau = (1-\tau)a + \tau\epsilon, \quad v^*(x_\tau, \tau, c) = \epsilon - a, \qquad \mathcal{L}_{\mathrm{FM}} = \mathbb{E}\|v_\theta(x_\tau, \tau, c) - (\epsilon - a)\|^2.$$

Sampling starts at $x_1 \sim \mathcal{N}(0, I)$ and integrates backward. **The current visual/language condition usually stays fixed across integration steps for one action request**, enabling prefix prefill and KV reuse while the action suffix changes. After a new camera observation, some or all cached representations must be refreshed. An unchanged language instruction does not guarantee that a multimodal prefix is unchanged after bidirectional processing.[1]

5.3 WAM: Autoregressive History Is Not a Frozen Prediction

An autoregressive WAM may cache confirmed history while jointly denoising the next predicted video-and-action chunk. After executing a chunk, DreamZero **updates its historical visual context with real observations**, reducing the compounding of open-loop video prediction errors.[11] Implementations must respect timestamps, masks, and dependency invalidation: if the observed world differs from the previously predicted world, the affected cached representations require updating or recomputation.

DiT caching is different. DreamZero exploits similarity in velocity directions across adjacent flow-matching steps; if a similarity criterion is satisfied, some velocity computation is reused. The paper reports an effective reduction from 16 to approximately four DiT passes in its setting. This is **approximate cross-denoising-step reuse**, unlike strictly valid causal KV reuse.[11]

6. Few-Step Flow Matching and WAM Inference

6.1 Two Distinct Inference Cost Profiles

For a VLA dominated by an action expert, an approximate latency profile is

$$T_{\text{VLA}} \approx T_{\text{encode}} + T_{\text{prefill}} + N_a T_{\text{expert}} + T_{\text{overhead}}.$$

For a WAM with a joint video-action DiT, it may be closer to

$$T_{\text{WAM}} \approx T_{\text{encode}} + T_{\text{history}} + N_{\text{eff}} T_{\text{DiT}}(L_{\text{video}} + L_{\text{action}}) + T_{\text{overhead}}.$$

Both are **diagnostic approximations**, not assumptions of strictly serial execution. Shortening the sampling trajectory, reducing video context, accelerating a DiT pass, and removing CPU-GPU synchronization can all help. Cutting N by a factor of four does not necessarily cut the complete request time by four.

6.2 DreamZero: System-, Implementation-, and Model-Level Improvements

DreamZero uses a 14B-parameter pretrained video-diffusion backbone to jointly generate future video and actions. Its paper reports approximately **5.7 s/chunk** for a naive setup. A combined optimization stack—two-GPU CFG parallelism, DiT caching, `torch.compile` and CUDA Graphs, kernel/scheduler improvements, quantization, and DreamZero-Flash—reaches **approximately 150 ms/chunk, a cumulative 38× speedup on the specified GB200 setup, and roughly 7 Hz closed-loop action updates**.[11,12]

Optimization level	Mechanism	Role in DreamZero
System: CFG parallelism	Separate conditional and unconditional branches across GPUs	Lower denoising-step wall time
System: DiT caching	Approximately reuse velocity predictions between steps	Fewer effective DiT calls
Implementation: compilation/CUDA Graphs	Reduce Python and launch overhead	Less CPU scheduling delay
Implementation: mixed precision/quantization	Reduce memory and arithmetic costs	Requires numerical and policy validation
Model: DreamZero-Flash	Adjust the joint video/action training noise distribution	More reliable few-step actions

Metric discipline. The **38×** figure comes from the cumulative, hardware-dependent stack in paper Table 1; it is not the contribution of Flash training in isolation. The separate **2.33×** figure in Table 3 compares 350 ms with 150 ms on a specific task. Those numbers must not be conflated.

6.3 DreamZero-Flash: Why Not Simply Reduce Four Steps to One?

With a shared video-action denoising time t, training primarily exposes the model to modalities at similar noise levels. But one-step inference may require **nearly clean motor actions even when the predicted video remains noisy**. This creates a train/inference distribution mismatch.

During its final training stage, DreamZero-Flash decouples the time distributions. One configuration in the paper is

$$t_{\text{video}} = 1 - \eta, \quad \eta \sim \text{Beta}(7, 1), \qquad t_{\text{action}} \sim \mathcal{U}(0, 1), \qquad \mathbb{E}[t_{\text{video}}] = 0.125.$$

Time-parameter convention matters: DreamZero uses $t = 0$ for the high-noise endpoint, **opposite** to the $\pi 0$ notation used in Section 5.2. Biasing video t toward zero therefore teaches the model to recover useful actions despite noisy video

latents. It does not simply disable the video branch: world prediction still contributes training supervision.[11]

Method (paper Table 3)	Denoising steps	Table-bussing task progress	Latency per chunk
DreamZero	4	$83\% \pm 6.1\%$	350 ms
DreamZero (naive one step)	1	$52\% \pm 10.2\%$	150 ms
DreamZero-Flash	1	$74\% \pm 10.1\%$	150 ms

These are **task-progress results (mean ± standard error) on the evaluated table-bussing task**, not general multi-task success rates. The 150 ms latency is not a universal promise for any accelerator, resolution, or workload. The key result is that **training-distribution design can improve the speed–action-quality tradeoff more effectively than merely shortening a sampler**.

7. Real-Time Robots: Asynchronous Closed-Loop Execution

7.1 Servo Rate versus Action Horizon

If the low-level controller runs at f_c and a chunk contains H control steps, its nominal duration is

$$T_H = \frac{H}{f_c}.$$

For example, one DreamZero bimanual setting uses 48 actions at 30 Hz, yielding a **1.6 s** action horizon. This duration is **not** the model replanning interval or the closed-loop reaction rate.[11]

Synchronous generation risks stalls at chunk boundaries. **Naive asynchronous execution** overlaps computation and execution, but can introduce discontinuities when switching chunks. **Real-Time Chunking (RTC)** conditions the new chunk on already committed actions, reducing action-switch transients by inpainting or constraining the generated continuation. Conceptually,

$$\tilde{a}_{0:k-1} = a^{\text{committed}}_{0:k-1}, \qquad \tilde{a}_{k:H-1} \sim p_\theta(\cdot \mid o_t, a^{\text{committed}}_{0:k-1}).$$

This is a conceptual continuity condition, not the complete RTC algorithm. Original RTC does not necessarily make the underlying Transformer faster. Training-time action-prefix conditioning is a related but distinct development.[2,3,8]

7.2 WAM: Correcting the World History Asynchronously

WAMs must distinguish **predicted future** from **observed reality**. While the robot executes the current chunk, the model computes the next one. After receiving timestamped camera/state feedback, the historical context is refreshed so subsequent predictions do not keep trusting already falsified visual futures. This mechanism can complement RTC, but **DreamZero's asynchronous controller must not be described as automatically implementing RTC**.[11]

If an action buffer contains R remaining control steps, a conservative scheduling check is

$$T^{\text{p99}}_{\text{infer}} + T_{\text{network}} + T_{\text{jitter}} + T_{\text{margin}} < \frac{R}{f_c}.$$

This is a **buffer-deadline engineering criterion**, not a proof of control stability or contact safety. Deployments should monitor observation age (Age of Information), p95/p99 latency, velocity/acceleration continuity at chunk transitions, tracking error, and safe fallback after timeouts.

7.3 The Additional Difficulty of Cloud WAMs

A remote WAM can centralize costly accelerators, but video upload, model queues, downlink actions, and network jitter all enter the effective feedback loop. The system should specify whether compressed frames or latent features are transmitted, how robot and server timestamps are reconciled, when stale replies are discarded, and whether missed deadlines trigger slowing, holding, or a controlled safe stop. **Server-side model throughput is not an adequate substitute for end-to-end latency distributions**.

For insertion, tool rotation, high-rate feedback, collision handling, and torque/impedance constraints should remain in suitable local control and safety layers. A cloud WAM should not be expected to serve as a high-bandwidth force-stability controller by itself.

8. Comparing π0.5, GR00T, DreamZero, and Dream-1

8.1 Optimization Priorities Vary by Model Family

Model family	Generated output	Main optimization priorities	Deployment challenge
$\pi0/\pi0.5$-style VLA	Flow-matching action chunks	Prefix KV, solver steps, RTC	Image encoding and chunk transitions
GR00T N1-style VLA	VLM-conditioned DiT action chunks	SDPA, DiT scheduling, image pipeline	Action latency, memory, interfaces
Autoregressive world model	Future state/vision tokens	Causal KV, history length, token reduction	History cache and drift
DreamZero-like joint WAM	Future video latents and actions	Video tokens, DiT caching, Flash, async execution	Video generation and system complexity

A blanket claim that "WAMs are always more expensive than VLAs" would be incorrect. A WAM can use latent states, low-resolution predictions, or a smaller network. Meaningful comparisons must fix the task, control requirements, hardware, and quality target.[7,9,11]

8.2 Dream-1: Cloud Service Pricing Is Not Local GPU Cost

Dream Labs' October 2026 public introduction describes Dream-1 as an API-accessible WAM that can operate previously unseen robot embodiments. Its advertised **service prices** are **US$3/hour for Dream-1** and **US$2/hour for Dream-1-flash**. The announcement also states that **each call returns 1.6 s of actions**.[13]

Those figures describe **published service billing and returned action horizon**, not a disclosed accelerator count, training budget, actual network latency per call, or guaranteed local deployment cost. In particular, a 1.6-second action horizon **does not imply a 1.6-second inference time**. Remote deployments still need independent p99 API latency, queueing, video uplink, and robot-interface measurements.

8.3 What Should a VLA–WAM Benchmark Report?

At minimum: accelerator model and count; precision/quantization; camera count and resolution; observation/prediction horizon; denoising steps; action-chunk length; actual model latency (preferably p50/p95/p99); network latency; model output rate; low-level execution rate; and success/task-progress metrics on the **same** tasks. **Tokens/s, parameter count, or hourly cloud pricing alone cannot determine suitability for contact-rich manipulation**.

9. Practical Priorities: Diagnose Before Optimizing

Measured bottleneck	First interventions	Poor first step
Training OOM	BF16, shorter video context, checkpointing, PEFT/FSDP	Write custom CUDA kernels immediately
Low GPU utilization	Video decoding, VAE caching, prefetch, synchronization	Add GPUs without profiling
Attention dominates	SDPA/FlashAttention, mask review, token reduction	Assume every mask supports the fastest kernel
VLA action generation is slow	Prefix KV, fewer solver steps, distillation, compilation	Reduce steps without checking contact success
WAM video-action DiT dominates	Fewer video tokens, DiT caching, few-step training, quantization	Confuse KV with DiT caching
Motion stutters despite fast inference	Async execution, RTC, synchronization, action-buffer slack	Continue focusing only on model FLOPs
Cloud requests occasionally time out	End-to-end p99, closer serving, stale-reply rejection, local fallback	Measure only average server latency
Precision contact is unstable	Fresh feedback, impedance/force control, safety constraints	Treat WAM/RTC as a safety controller

Recommended priority order: first make execution continuous and safe; next profile full observation-to-action latency; finally optimize attention, joint DiT passes, samplers, or communication according to evidence. The attraction of WAMs is a learned prior over world evolution, not immunity from timing, contact, and systems engineering.

10. Six Common Confusions

Q1. Is FlashAttention the same as DreamZero-Flash? No. FlashAttention reduces IO within a given attention operation. DreamZero-Flash changes the training noise distribution of a joint video-action generator to improve few-step action quality.

Q2. Are KV caching and DiT caching interchangeable? No. KV caching can exactly retain stable historical context when dependencies permit. DiT caching generally reuses approximate predictions or features across adjacent sampling steps.

Q3. Must a WAM decode predicted video into RGB to control the robot? No. Joint generation can occur in a latent space. Skipping pixel rendering does not necessarily eliminate the cost of predicting video latents.

Q4. Is DreamZero's 38× improvement solely due to one-step generation? No. It is a cumulative systems-and-model improvement. The separate four-step-to-one-step comparison in Table 3 is 2.33× with different task progress.

Q5. Does 150 ms/chunk imply a 30 Hz policy feedback loop? No. The model's chunk-production rate differs from low-level action execution frequency. Observation updates, timestamps, and scheduling determine reactivity.

Q6. Must a VLA/WAM researcher start by mastering CUDA and implementing KV caches? No. First learn how to characterize workloads, verify cache validity, budget VRAM, measure latency, and retain local control safeguards. Triton/CUDA becomes important when profiling establishes a kernel bottleneck.

11. Conclusion

Efficiency in embodied foundation models is not one latency number. It jointly concerns **training compute**, **model delay**, **observation freshness**, **action continuity**, **task success**, **and serving cost**. VLAs often concentrate on conditioning-prefix reuse and fast action-expert generation. WAMs additionally manage video-token budgets, joint-DiT computation, and the correction of predicted world history.

In one sentence: FlashAttention improves **one** attention pass; KV caching avoids recomputing an unchanged prefix; DiT caching approximates reuse **across denoising steps**; DreamZero-Flash trains for **reliable few-step actions**; RTC and asynchronous execution connect slower models to continuously moving robots; and high-rate feedback and safety remain the responsibility of local control layers.

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